

GitHub Actions for Automated Email Updates

TL;DR: Save time, make friends, get more sleep

Megan Holtorf

Cascadia R Conference 2025



The story of a running club...



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Mark Zuckerberg's Profile

Puget Sound

quick search**go**[My Profile \[edit \]](#)[My Friends](#)[My Parties](#)[My Messages](#)[My Account](#)[My Privacy](#)

Picture

[Send Mark a Message](#)[Poke Him!](#)

Connection

This is you.

Mutual Friends

You have 19 friends in common with Mark

Information

Account Info:

Name: Mark Zuckerberg
Member Since: The beginning (February 4, 2004)
Last Update: July 19, 2005

Basic Info:

School: Harvard
Status: Student
Sex: Male
Residence: Todd 311
Birthday: May 14, 1984
Home Town: Dobbs Ferry, NY
High School: Phillips Exeter Academy

Contact Info:

Email: mzuckerb@fas.harvard.edu
Screename: zberg02
Mobile:

+ Create

June 2025

<

>

M	T	W	T	F	S	S
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6

👤 Search for people

My calendars



Other calendars



MON
26

TUE
27

WED
28

THU
29

FRI
30

SAT
31

SUN
Jun 1

Memorial Day

2

3

4

5

6

7

8

9

10

11

16

17

18

23

24

25

26

27

28

29

30

Jul 1

2

3

4

5

6

Junet

Independence Day

8am Fourth of July Free

Father's Day

15

22

10:30am Post Cascadia

9am Rose Festival River

Post CascadiaR Presenter Run

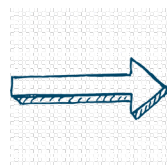
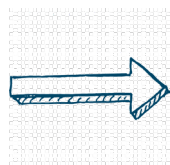
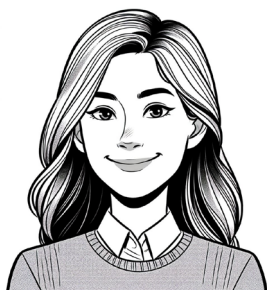
Sunday, June 22 · 10:30am – 12:30pm

📍 South Park Blocks

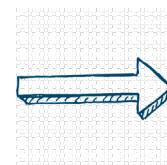
☰ Explore Portland and meet fellow R Enthusiasts.

📅 Running Events
Created by: Megan Holtorf

What if we just sent everyone an email?



GitHub Actions



01 Encourage sign-ups for a Google group

*optional but highly recommended

02 Pull events from the calendar with {ical}

03 Check for new updates or upcoming events

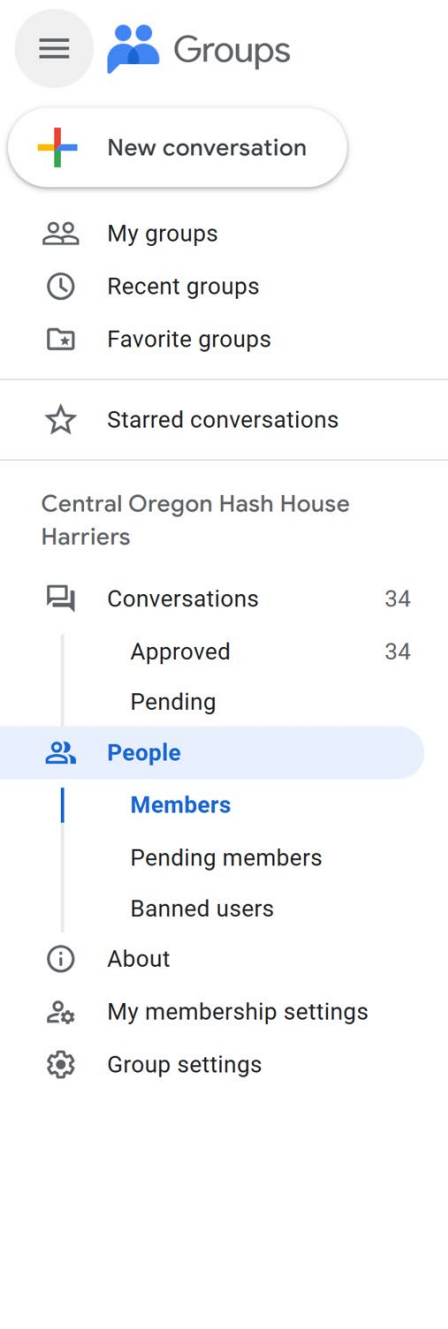
04 Categorize the events

05 Format the results in {quarto}

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← Settings

General

Add calendar

Import & export

Settings for my calendars

Calendar

Running Events

Calendar settings

Auto-accept invitations

Public URL to this calendar

https://calendar.google.com/calendar/embed?src=9c1276ad221387cb8630d6642f1d9cf0244

Use this URL to access this calendar from a web browser.

Embed code

<iframe src="https://calendar.google.com/calendar/embed?src=9c1276ad221387cb8630d6642f1d9cf02449d7f16e"

Use this code to embed this calendar in a web page.

You can customize the code or embed multiple calendars.

Customize

Public address in iCal format

https://calendar.google.com/calendar/ical/9c1276ad221387cb8630d6642f1d9cf02449d7f16e

Use this address to access this calendar from other applications

```
events <-  
ical::ical_parse_df(  
  "https://calendar.google.com/calendar/ical..."
```

	uid	summary	dtstart	dtend	description	location
1	CSVConvert00ddbe519b19969fab91d1cc896582df	Portland Pet & Pedal Run	2026-05-03 10:00:00	2026-05-03 12:00:00	A fun run where participants are encouraged to bring their ...	Laurelhurst Park
2	CSVConvert025c649f5bd9df8bdb0ad27608319ee3	Rose Festival River Run	2026-06-29 09:00:00	2026-06-29 11:00:00	Join us for a floral-inspired run as part of Portland's Rose Fe...	Willamette River Trail
3	CSVConvert0372d5a71f107dd7a700ad3a70712c03	Shamrock Stride	2027-03-16 09:00:00	2027-03-16 11:00:00	A St. Patrick's Day themed run followed by a social at a local...	Downtown Portland
4	CSVConvert0545a6e16519ee4256827347b20cf374	Harvest Festival Fun Run	2026-10-19 10:00:00	2026-10-19 12:00:00	A family-friendly run amidst the harvest festival activities on...	Kruger's Farm
5	CSVConvert084f4d78a125400384a66e70a4909c50	Leap Year Run	2026-02-24 08:30:00	2026-02-24 10:30:00	Celebrate the leap year with an invigorating run through M...	Mount Tabor Park
6	CSVConvert08a865201da35cf7b5fb5eae552f236	Halloween Costume Run	2025-10-26 18:00:00	2025-10-26 20:00:00	Dress up and enjoy a spooky run along the river, followed b...	Portland Waterfront
7	CSVConvert0ba8850e808b607b9c9a55e56b88a1b0	Halloween Costume Run	2026-10-26 18:00:00	2026-10-26 20:00:00	Dress up and enjoy a spooky run along the river, followed b...	Portland Waterfront
8	CSVConvert0e87596c57b629bdc355ce5963c9c65f	Autumn Equinox Trail Challenge	2025-09-21 08:30:00	2025-09-21 10:30:00	Welcome fall with a challenging trail run full of autumn scen...	Powell Butte Nature Park
9	CSVConvert11e40eab1c63294b189e8cce03a43acf	Brewery Dash & Social	2026-01-27 15:00:00	2026-01-27 17:00:00	A short run followed by a brewery tour and tasting session.	Deschutes Brewery
10	CSVConvert14952dd722407f4d20f5b6c59d4a67a9	April Showers Spring Run	2027-04-12 08:00:00	2027-04-12 10:00:00	Embrace the spring showers with a refreshing run through b...	Peninsula Park
11	CSVConvert1868b152293fbaad07b7d181c71b0cb8	Summer Brewery Crawl & Run	2026-08-17 16:00:00	2026-08-17 18:00:00	A fun run with stops at some of Portland's best craft breweri...	Various Downtown Breweries
12	CSVConvert1e97de47e0b9fe63906737762c356756	Pi Day 3.14 Mile Race	2027-03-14 15:15:00	2027-03-14 17:15:00	Celebrate Pi Day with a special 3.14-mile race, followed by p...	Eastbank Esplanade
13	CSVConvert259e44f91ed53b1ea2cabb45efcca539	Turkey Trot & Feast	2026-11-28 08:00:00	2026-11-28 10:00:00	Celebrate Thanksgiving with a morning run and post-race fe...	Sellwood Park
14	CSVConvert29c845c9b302b8a69a0e78cccd3d2474	Forest Park Fun Run	2027-01-13 09:00:00	2027-01-13 11:00:00	Kick off the year with a scenic run through Forest Park's bea...	Forest Park
15	CSVConvert2ab73ad37457b3233be6167069aede62	Brewery Dash & Social	2027-01-27 15:00:00	2027-01-27 17:00:00	A short run followed by a brewery tour and tasting session.	Deschutes Brewery
16	CSVConvert2d3b47f47313d10e8d3e43d81807a548	Mother's Day Brunch & Run	2026-05-12 09:00:00	2026-05-12 11:00:00	Celebrate mothers with a lovely morning run followed by a ...	Laurelhurst Park
17	CSVConvert2de29db91e1515f777e0b4db7f19006e	Pi Day 3.14 Mile Race	2026-03-14 15:15:00	2026-03-14 17:15:00	Celebrate Pi Day with a special 3.14-mile race, followed by p...	Eastbank Esplanade
18	CSVConvert34b0ee514a7adb78e5531c3983619aa	Trail Mix Challenge	2026-04-06 09:00:00	2026-04-06 11:00:00	Tackle a variety of trails and enjoy post-run snacks inspired ...	Hoyt Arboretum
19	CSVConvert3b33ea44e48d53056b2c80398592b7a2	Winter Solstice Run	2026-12-21 07:00:00	2026-12-21 09:00:00	Embrace the shortest day of the year with a mystical early m...	Forest Park
20	CSVConvert3ba8b62f1bf8a00a84d0b824679e255f	Turkey Trot & Feast	2025-11-28 08:00:00	2025-11-28 10:00:00	Celebrate Thanksgiving with a morning run and post-race fe...	Sellwood Park
21	CSVConvert3f5cf64d0ce7b074b1bff6e4a36acff	Harvest Festival Fun Run	2025-10-19 10:00:00	2025-10-19 12:00:00	A family-friendly run amidst the harvest festival activities on...	Kruger's Farm
22	CSVConvert4044c690490c0cb8a61f5f75879d2b777	Super Bowl Sunday Sprints	2027-02-02 08:00:00	2027-02-02 10:00:00	Get your heart racing with short sprints before settling in for...	Providence Park
23	CSVConvert4be1837d8c3cd666ba46f26c057de7c5	Portland Parks Challenge	2026-07-12 08:30:00	2026-07-12 10:30:00	A fun challenge to run in multiple parks across Portland in o...	Various Parks

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02 Pull events from the calendar with {ical}

03 Check for new updates or upcoming events

04 Categorize the events

05 Format the results in {quarto}

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07 Schedule it in GitHub actions

08 Rest up for your next long run!

```
events <-
  events %>%
  mutate(description = gsub("\n", "<br>", description),
         dtstart =
           if_else(hour(dtstart) == 0 & hour(dtend) == 0,
                   # convert dtstart from GMT to PST or PDT
                   lubridate::force_tz(dtstart, tzone = "America/Los_Angeles"),
                   dtstart),
         daysuntill = as.numeric(difftime(as.Date(dtstart),
                                           as.Date(Sys.Date()),
                                           units = "days"))) %>%

  arrange(dtstart)

# Identify events triggering in 10 days
trigger_10 <-
  events %>%
  filter(daysuntill == 10)

# Identify events triggering in 2 days
trigger_2 <-
  events %>%
  filter(daysuntill == 2)
```


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```
# Identify events with updates:
# - Upcoming events within the next 2 days
# - Events with new updates in the last 24 hours
updates <-
  events %>%
    filter(dtstart > Sys.Date(),
           daysuntill < 2 &
           difftime(last.modified, Sys.Date(), units = "hours") > -24)
```

```
# Identify events happening next two days:
# - Future events within the next 2 days
# - Exclude events already marked with updates
twoday <-
  events %>%
    filter(as.Date(dtstart) > Sys.Date() &
           daysuntill == 2 &
           !uid %in% updates$uid)
```

```
# Identify events happening next week:
# - Future events within the next 10 days
# - Exclude events already marked with updates
nextweek <-
  events %>%
    filter(as.Date(dtstart) > Sys.Date() &
           daysuntill <= 10 &
           !uid %in% updates$uid &
           !uid %in% twoday$uid)
```

```
# Identify events happening 10 days from now
upcoming <-
  events %>%
    filter(dtstart > Sys.Date() + 10 &
           dtstart < Sys.Date() + 60 &
           !uid %in% nextweek$uid &
           !uid %in% twoday$uid)
```

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```
---
title: "Hoppy Trails Running Club"
output: blastula::blastula_email
editor: visual
---
```

Hoppy Trails Running Club

 Coming soon...

Summer Solstice Evening Run

Date: Saturday June 21

Time: 07:00 PM

Where: Eastbank Esplanade

Celebrate the longest day of the year with a sunset run along the river.

Post CascadiaR Presenter Run

Date: Sunday June 22

Time: 10:30 AM

Where: South Park Blocks

Explore Portland and meet fellow R Enthusiasts.

Save the Date!

Get this events on your calendar!

Rose Festival River Run

Date: Sunday June 29

Time: 09:00 AM

Where: Willamette River Trail

Join us for a floral-inspired run as part of Portland's Rose Festival.

For up to the minute event details, add the Group Runs calendar to your own calendar using [Google](#) or [iCal](#) for iOS users.

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```
library("blastula")
```

```
sender_email <- "youremailhere@gmail.com"  
reply_email <- "hoppytrails@googlegroups.com"  
distribution_email <- "hoppytrails@googlegroups.com"
```

```
# Generate the email object using blastula  
email_object <- blastula::render_email(here::here("R",  
"email_content.qmd"))
```

```
# Send email if there's an even next week or any new updates  
if (nrow(trigger_2) + nrow(trigger_10) + nrow(updates) >= 1)  
  blastula::smtp_send(  
    email_object,  
    from = c("Hoppy Trails" = sender_email),  
    to = reply_email, # Update to the reply email or distribution  
    email  
    bcc = c(distribution_email, sender_email), # include the user,  
    Google groups won't send directly to sender.  
    subject = paste0("Hoppy Trails Running Club"),  
    credentials = creds_envvar(  
      user = sender_email, # Update if transferred to new owner  
      host = "smtp.gmail.com",  
      port = 465,  
      use_ssl = TRUE,  
      pass_envvar = "GMAIL_APP_PASSWORD"  
    ) # Points to a GitHub secret with Google App password, update is  
    owner changes  
  )
```

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```
on:
  schedule:
    - cron: '0 16 * * *' # Run every day at 4pm UTC
jobs:
  run_email_sender:
    runs-on: ubuntu-latest

    steps:
      - name: Check out repository
        uses: actions/checkout@v3

      - name: Set up R
        uses: r-lib/actions/setup-r@v2
        with:
          r-version: 'renv' # Use the version recorded in the renv.lock file

      - name: Set up Pandoc
        uses: r-lib/actions/setup-pandoc@v2

      - name: Cache packages
        uses: actions/cache@v3 # Cache packages
        with:
          path: ~/.local/share/renv
          key: ${{ runner.os }}-renv-${{ hashFiles('**/renv.lock') }}
          restore-keys: |
            ${{ runner.os }}-renv-

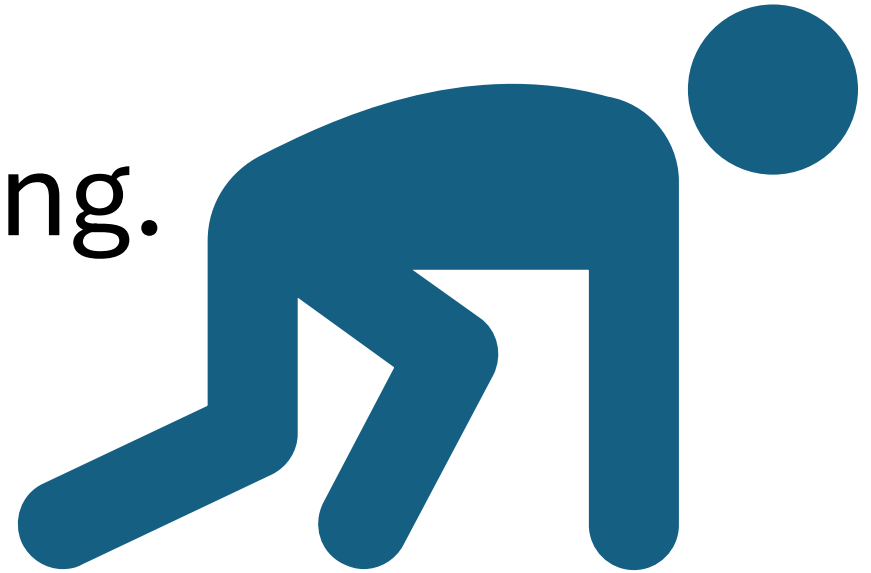
      - name: Setup renv # Set up R env to install packages
        run: |
          R -e 'install.packages("renv")'
          R -e 'renv::restore()'

      - name: Render email
        run: |
          source("R/send_email.R")
        shell: Rscript {0}
        env:
          GMAIL_APP_PASSWORD: ${{ secrets.GMAIL_APP_PASSWORD }}
```


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Cool.... But I hate running.



Users are everywhere

Your closest friends can make for **great test subjects**. They can be even more critical and demanding, but your job isn't on the line.

Adaptation is crucial

Are voicemail services still a thing? Yahoo groups sure isn't. This process will also change one day. **What comes next?**

Automation can free up time

But it's not fool proof or maintenance free. Sometimes APIs change, data gets formatted weird. And sometimes **it just fails**.

Find low-risk learning opportunities

Trying something new in a low-risk environment helps prepare you for the high-stakes projects. **"You won't know until you try"**

**Take my
picture for
code and
instructions**

